

What a Streaming Decoder Costs

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What a Streaming Decoder Costs: A Five-Gate Benchmark of Twenty Neural Audio Decoders, and a Metric That Failed Its Own Validity Check

Panagiotis Gkilis · BedVibe Studios, Oslo ORCID 0009-0007-3805-170X

Version 1.0.1 — corrected. Supersedes v1.0.0 (DOI 10.5281/zenodo.22798416), which remains permanently available. An independent post-publication adversarial review found

reporting and public-reproducibility defects; `CORRECTIONS_v1.0.1.md` lists every one, with the original wording beside the corrected wording. **No measurement changed** — the corrections are to prose, to release packaging and to claims that outran their evidence. Publication manifest `1e93e28b9adb3b1f`. Every numeric value is generated from a frozen artifact by `make_paper_tables.py`; the prose cites the generated tables and adds no number of its own.

Abstract

Choosing a neural audio decoder for a streaming text-to-speech product requires four separate answers — does it run, does it truly stream, how fast is it, and what does it cost the signal — and the literature answers them with figures that are rarely comparable. We benchmarked **twenty decoder arms** under one frozen protocol on identical audio, with each gate’s protocol pre-registered before it was measured and each instrument attacked before it was trusted. Two later freezes — Gate 4’s analysis layer, and revisions 4–6 of the Q5 specification — were **post-measurement repairs** rather than prospective pre-registrations; §3 and §9 say which, why, and what changed as a result.

Three results stand out, and two of them are negative.

First, “streaming” is a property almost nothing in the candidate set has. Of twenty arms, only **three** reproduce full-context output from partial input while carrying state — all three explicitly causal FocalCodec configurations. **Fifteen** use stateless chunking rather than load-bearing streaming state. One streams its network but never emits audio from partial input, and its label says so. Several of the fifteen never claimed to stream; the classification is a measurement of what they do, not a charge against what they advertised.

Second, our sole pre-registered cross-arm quality metric failed the validity instrument we built to catch exactly that. Griffin-Lim, a zero-parameter phase-retrieval algorithm from 1984, does not merely score close to the trained mel decoders under our mel-cepstral distance — it scores **best in six states of six**. The pre-registration declared in advance what that would mean, so the finding is reported as a property of the metric, not as a decoder ranking.

Third, a second measurement with no shared code path reached the same verdict on the same three arms. Measuring speaker-embedding retention with three separately calibrated speaker encoders, the three causal configurations show **near-zero median additional change** when streamed, under the tested encoders, supported recordings and imposed chunking regime: offline-vs-streamed cosine **0.999866–**

0.999946, and a paired retention delta of -3.9×10^{-5} to -5.3×10^{-6} — small, and strictly negative. The other thirteen lose between **-0.078** and **-0.742** *on the retention delta*; that is a different quantity from the cosine and the two are never quoted as one range. ⚠️ The two measurements are independent in *construction* — a spectral distance and three speaker encoders, sharing no implementation — but **not** in *data*: both run on the same reconstructions of the same corpus, so a corpus-level or reconstruction-level artefact would move both. What their agreement adds is that two different representations of the signal register the same three survivors.

We also report what we could not establish: perceptual quality was never measured, and the identity result is a statement about speaker-encoder representations rather than about how the audio sounds. The study went through **six rounds of independent adversarial audit**: four before publication, which found three defects that changed conclusions; a fifth against the published package, which found that its advertised reproduction commands did not run and that four claims outran their evidence; and a sixth against the correction itself. All are reported here, and in `CORRECTIONS_v1.0.1.md`, because the repairs changed what the paper is entitled to say.

1. Introduction

A team building a streaming TTS system has to choose a decoder. The published record makes that harder than it should be. Real-time factors are quoted without hardware; “streaming” is claimed for architectures that require the whole utterance; quality figures come from different corpora at different sample rates through different measurement paths. None of it composes into a decision.

This benchmark exists to make one decision inspectable rather than asserted. It measures **representation** → **waveform only**. Generator and autoregressive timing are outside its scope, and a decode-only number read as end-to-end TTS performance would be a failed report.

The study is organised as five gates, each pre-registered before it was run:

gate	question
0 — candidate	which decoders are runnable, licensed and comparable at all?
1 — functionality	does the arm produce structurally valid audio?
2 — streaming	does it <i>truly</i> stream, or is it stateless chunking?
3 — performance	latency, real-time factor, first-audio, jitter, memory

gate	question
4 — quality	what does the round trip cost the signal?
Q5 — identity	does the speaker survive, and does he stay one person?

One methodological commitment shapes everything below: **an instrument is attacked before it is trusted, and its author does not certify it.** Two of the three headline findings are consequences of that rule rather than of the decoders.

2. Candidate set, and a reframing that changed it

The candidate gate closed with a finding that changed the population: **for a TTS build, only the decoder needs to be causal.** The encoder runs at training time on complete utterances; at inference the generator emits tokens directly and no encoder sits in the serving path. NVIDIA’s NanoCodec makes the point concrete — non-causal encoder, causal HiFi-GAN decoder. “Is this codec causal?” is therefore the wrong question, and asking it nearly excluded several candidates for the wrong reason.

The measured population is twenty arms spanning two routes — **Route A**, representation → waveform through a codec’s own decoder, and **Route B**, mel → waveform — at three sample-rate bands (16 kHz, 22–24 kHz, 48 kHz), because sample rate is an architectural variable and was never normalised away.

Two arms carry declared non-results rather than numbers. `nanocodec` is **BLOCKED — PLATFORM**: it could not be loaded here, and no timing figure is manufactured for it. `melflow` is AGPL-3.0 and carries a label on every result it produces: research evidence, not a shippable closed-product dependency. Its released streaming path also **did not execute as-is** and required two structural repairs, which an independent delta audit confirmed were inert — touching no weight and no arithmetic operation.

3. Method

Corpus. Seven speakers, English, one studio, one recording chain, six emotional states — Neutral, Angry, Happy, Scared, Shouting, Whisper. **1,512 distinct source recordings**, every file hashed with SHA-256 before use: Set S (1,302 recordings, 31 texts, all six states, 2.54–13.50 s), Set L (210 recordings, 15 long texts, Neutral and Whisper, measured on a derived 14 s top rung), and a calibration subset (126 recordings) held out by construction. The source audio is private, rights-cleared and consented, and is **not published**.

Environment. NVIDIA GeForce RTX 5080, torch 2.10.0+cu128, CUDA 12.8, seed 20260911, eager mode throughout — no `torch.compile`, no CUDA graphs. Timing used 30 warm repetitions plus 3 cold, 5 warm-ups discarded, block-randomised, one GPU process at a time.

Freeze discipline. Every gate’s apparatus is frozen and hashed, `--verify` reports drift, drift was 0 at every checkpoint, and superseded revisions are retained rather than discarded. **Not every freeze is prospective, and the records say which are which** — this is stated here because an earlier version of this paper claimed uniformly that each apparatus was frozen before the data it governs existed, which the freeze records do not support.

The protocol and the gate definitions were written before any decoder ran.

`GATE4_FREEZE.json` seals the Gate 4 corpus, instrument and selection rule and records that at that moment “no decoder has run and no quality number exists”. `GATE3_FREEZE.json` is a different kind of record: it seals Gate 3 on completion, after its measurements existed. Gate 4’s **analysis** layer was likewise frozen once quality numbers existed, and its own record says so without hedging — freezing “does NOT make them blind, and nothing later can”.

The Q5 specification went through six revisions. Revisions 1–3 were frozen with no Q5 artifact of any kind on disk. **Revisions 4–6 were post-measurement repairs:** measured artifacts already existed when each was frozen, and each carries `OVERRIDDEN` — artifacts existed at spec-freeze time together with the sentence “no claim of blindness is made for this revision”. The guards record `Q5_CELLS.jsonl`, `Q5_EMBEDDINGS.npz`, `Q5_CALIBRATION.json` and `Q5_FUZZ.json` present at all three, and by revision 6 the aggregated `Q5_RESULTS.json` as well. **No blindness is claimed for any of them.** What they changed, and what changed in the labels as a result, is in §9. What freezing establishes throughout is that the decisions were declared, fixed and auditable; it does not establish that the analyst was blind to the data, and for revisions 4–6 they demonstrably were not.

Statistics. The experimental unit is the **recording**, never the cell. Intervals are 95% percentile intervals from a cluster bootstrap over recordings, 4,000 resamples, seed 20260911, aggregated within (speaker × state) before across. **No hypothesis tests are performed and no p-value is reported** — intervals and effect sizes only.

4. Gate 1 — functionality

Every arm passes a structural gate before any claim: finite samples, peak in (0, 1], duration within one frame of that arm’s own rate, energy ratio 0.25–4.0, float32, correct channel count. **Nineteen of twenty arms PASS.** The twentieth is `nanocodec`, recorded as `NOT RUN` — arm could not be loaded.

Gate 1 is deliberately weak. It establishes that a number *may* be computed, not that it means anything.

5. Gate 2 — true streaming versus stateless chunking

This is the gate we expected to be routine, and it was the most discriminating of the five.

An arm is `TRUE_INCREMENTAL` only if **both** conjuncts hold: **C1**, stateful chunked decode reproduces full-context output to $\max|\text{err}| \leq 1\text{e-}3$; and **C2**, state is load-bearing — the stateless error is at least 10× the stateful error. C2 exists because C1 alone can be satisfied by an arm that ignores state.

classification	arms
<code>STATELESS_CHUNKING</code>	15
<code>TRUE_INCREMENTAL</code> (representation → PCM)	3 — <code>focalcodec_50hz_2k_causal</code> , <code>focalcodec_50hz_4k_causal</code> , <code>focalcodec_50hz_65k_causal</code>
<code>TRUE_INCREMENTAL</code> — neural decoder only, iSTFT streaming not established	1 — <code>melflow</code>
<code>BLOCKED – PLATFORM</code>	1 — <code>nanocodec</code>

The three causal configurations pass C2 with ratios of **3,530 to 36,740** against a threshold of 10 — state is not marginally load-bearing, it is decisive.

The negative controls are what make the positives mean anything. The non-causal FocalCodec configurations are driven through the *identical* stateful code path. Their C2 ratio is **exactly 1**: carrying state changes nothing. Had a negative control passed, the experiment would have been void rather than the arm promoted.

`melflow` is the honest boundary case. Its neural decoder is driven frame-synchronously with carried state and passes both conjuncts, but the inverse STFT runs once over the assembled spectrogram, so **no PCM is emitted from partial input**. A control measured that the overlap-add stage is not free — a naive split-and-concatenate inverse STFT departs by 134.65% — so streaming the neural decoder does not imply streaming the chain. Its label carries the qualification and it is never counted with the arms that emit PCM.

We also measured per-chunk length drift rather than assuming it away: four arms accumulate a constant offset per chunk, worst case **47,360 samples ≈ 1.97 s** for `vocos_mel24` and `griffinlim`. Part of the error attributed to boundary damage in a naive analysis is misalignment.

6. Gate 3 — performance

One environment finding governs how these numbers may be read. A controlled probe — byte-identical tokens, same GPU, `encodec24_q8` as a single arm in every environment — found the incumbent’s environment **50.2% faster** than the reference (`transformers` 4.35.2 versus 4.57.3), a near-constant 3–4 ms device-side offset at identical torch. Host overhead is 0.022 ms versus 0.024 ms while CUDA time is 10.344 ms versus 6.328 ms: the two library versions launch different GPU work for the same decode. Consequently `fish_modified_dac` **may not be compared numerically against any arm measured in the reference environment.** Its numbers stand within its own environment. This says nothing about whether the incumbent is fast or slow; it says the comparison does not exist.

At the ~80 ms chunk anchor, **first-audio latency (TTFA) spans 1.37 ms** (`vocos_mel24`) **to 27.64 ms** (`bigvgan22`) across the arms that have a numerical TTFA at all. `melflow` is not in that range: its TTFA is **NOT ESTABLISHED**, because it emits spectrogram frames and producing PCM from a partial stream needs an overlap-add stage this study records as not established. Its **375.91 ms is a steady-state p50**, a different quantity, and the two are never mixed — in eager mode and therefore an upper bound, since its upstream recommends CUDA graphs. Real-time margin spans **64×** down to **0.2×**. Seventeen arms show zero underrun at the anchor; `qwen3_tts_tokenizer_12hz` shows 0.0013 with a p99 of 17.39 ms against a **max of 208.10 ms** — a tail that a p50 hides completely.

Speed does not imply validity, and Gate 3 refuses to let it. Two arms produce streamed output that fails the study’s own §8 validity gate at every chunk size tested: `vocos_mel24` (0/7 conditions) and `griffinlim` (0/6), both on `duration_ok`. A third, `focalcodec_12_5hz`, fails **at the anchor** but not everywhere: it passes **2 of 5** tested conditions, failing on `energy_ok` at the smaller sizes, so its minimum viable configuration is **640.8 ms**, not 80 ms. Their timing numbers are real; what they timed at the anchor is not a valid streaming configuration. `vocos_mel24` is the sharpest illustration: the **fastest** first-audio in the study, and no achievable streaming configuration at all.

7. Gate 4 — reconstruction quality, and a metric that failed

Gate 4 ranked on **one** metric, mel-cepstral distance, chosen after nine metric variants were implemented and attacked; the other eight were demoted to detectors, permitted to report that two signals differ but never that one decoder is better. Ranking required three conjuncts: the cell inside a validated support domain, a paired 95% recording-cluster interval excluding zero, and an effect exceeding a state-aware absolute floor.

Of 108 within-route pairs: **46 SEPARATION ESTABLISHED · 42 ORDERED, SEPARATION NOT ESTABLISHED · 20 QUALITY COMPARISON NOT ESTABLISHED**. Only **three** pairs are ROBUST, all of them statements that `mimi_q8` is worse than another arm, and one further pair is floor-fragile. Forty-two of the ordered pairs rest on **3 to 17 paired cells** and **all 42 contain a FocalCodec arm** — the survivors are that arm’s most favourable cells, so the direction survives its own selection but the magnitude does not.

7.1 Q9 fires

The pre-registration installed Griffin-Lim as a zero-parameter floor and stated in advance what it would mean if a 1984 phase-retrieval algorithm scored close to a trained neural codec: *a finding about the metric*.

It did not score close. It scored **best, in six states of six** (see T-Q9 below, generated): 81.6 versus 102.6 and 127.7 at Neutral, and the same ordering in every other state.

A proposed explanation was tested and **refuted**. The hypothesis that the metric’s low-level frame floor favours a magnitude-matching algorithm failed: splitting each cell at its own median energy, Griffin-Lim leads in **both** halves and by *more* in the loud half (21.4 versus 16.0).

What remains is an explanation **consistent with** the result rather than a demonstrated mechanism, and it is worth stating the difference. Mel-cepstral distance as implemented here is derived from the magnitude spectrum — mel filtering of the power spectrum, a logarithm, a cepstral transform, and removal of coefficient zero — and it is blind to phase by construction. Griffin-Lim iterates toward consistency of the **linear STFT magnitude** and optimises nothing else, accepting whatever phase error that leaves. The two objectives are aligned in domain but are **not the same quantity**, and this package does not demonstrate that Griffin-Lim minimises the released cepstral distance exactly. That alignment is the most plausible account of a zero-parameter algorithm winning six states of six; it is not established here, and no claim in this paper rests on it.

What is established is narrower and does not depend on the mechanism: a metric whose ranking puts an untrained phase-retrieval algorithm first is not, on its own, a sufficient authority for cross-decoder quality. That is the pre-registered consequence, and it fired on the evidence rather than on the explanation. It does not follow that optimising a related quantity automatically disqualifies a metric — the disqualifying observation here is the ranking itself. Two scope limits travel with this finding: the instrument exists only in Route B, so the same check is `NOT ESTABLISHED` — not “passed”

— in Route A; and no new metric was introduced after seeing the data, because choosing a ruler once you know who won under the old one is precisely the failure Q9 exists to catch.

7.2 What the support domain removed

Coverage varies by more than two orders of magnitude: `griffinlim` 99.3% of cells rankable, `mimi_q32` 79.4%, and every FocalCodec arm between **1.0% and 0.2%**. A FocalCodec quality comparison is therefore drawn from the few percent of cells where that arm retained broadband energy. This is stated rather than smoothed, and it is why 20 of the 20 refused pairs involve FocalCodec.

8. Q5 — speaker identity

Gate 4 pre-registered speaker-identity retention as **Q5** and marked it *metric not built, not fuzzed*. It was built afterwards as a separate layer, consuming the frozen reconstructions without reopening any Gate 4 verdict.

Q5 measures two quantities that a single similarity score conflates, and refuses to merge them:

- **Retention (R)** — cosine between a source recording and its own reconstruction.
- **Dispersion (D)** — cosine between *different* reconstructions of the same speaker, so a decoder that moves a voice consistently can be told apart from one that makes the speaker wander.

Both are read against two references measured on **source audio alone, before any reconstruction was opened**: the human within-speaker ceiling (how far apart two genuine recordings of one speaker sit) and the between-speaker floor.

The panel is three independently calibrated encoders (T1), chosen mechanically from a previously frozen fourteen-encoder study before any Q5 number existed. Cosine values are **encoder-specific and never averaged**; the panel's role is agreement, not a pooled score.

The instrument was calibrated and gated before use (T2, T3). Each encoder's usable range G — the distance between “same person again” and “a different person” on this corpus — is 0.2663 to 0.4056, and each encoder's response to a one-sample shift is 100× smaller than the bound that would demote it. All 24 separability gates pass.

8.1 Retention

Retention (T4) runs from **0.9952** (`griffinlim` , `ecapa`) down to **0.4316** (`focalcodec_12_5hz`). **The three encoders show broadly similar ordering, with some pairwise reversals** — `encodec24_q8` sits above `dualcodec_12hz_v1` under `ecapa` (0.8781 versus 0.8621) and below it under both ReDimNet encoders, and `encodec_vocos` leads `mimi_q32` under `ecapa` and `redimnet_M_vb2_ptn` but trails it under `redimnet_b6_lm` (0.8546 versus 0.8581). Per-encoder spread reaches 0.1193, which is why a bare cosine without its encoder name is not a result, and why neighbouring arms should not be read as ranked against each other at all.

Griffin-Lim has the highest speaker-embedding retention of the eighteen arms. That sentence is the whole claim. It is not a statement that Griffin-Lim is the best decoder, and it is not a perceptual claim — it is a measurement of how well one representation survives a round trip. Read together with Q9 it says something sharper: **two instruments, a spectral distance and three speaker encoders, both rank a phase-blind algorithm first.** Both are blind to the same thing, and only a listening test can settle what that costs.

8.2 Dispersion

Of 540 labelled strata (T5), the dominant outcome is `NOT ESTABLISHED` (314) — which here means dispersion **statistically indistinguishable from the human ceiling**, not absence of information. `INCONSISTENT / DRIFTING` accounts for 152, concentrated in the FocalCodec family; `PRESERVED + CONSISTENT` for 19; and `STABLE OFFSET` — voice moved but internally coherent — for 8, almost all of them FocalCodec at 25–65 Hz under `redimnet_M_vb2_ptn`. That last category is the one a single retention score cannot express.

One dispersion result survives its own confound and one does not. `mimi_q8` disperses *inside* the validated support domain (180 of 217 cells in support, D 0.6237 against a ceiling lower bound of 0.6440), so its drift is not a band-limiting artefact. Every FocalCodec drift verdict, by contrast, rests on out-of-support cells — **0 to 6 in-support cells per stratum, and 50 strata with zero** — so for that family identity dispersion and band-limiting **cannot be separated in this design**, and no causal claim is made.

A caveat measured on human speech alone: for `ecapa` and `redimnet_b6_lm` the *human* cross-state ceiling already sits near their own impostor floor. Every one of the 69 `APPROACHES IMPOSTOR RANGE` markers fell on those two encoders and on cross-state strata only. **That is an encoder baseline limitation, not decoder drift.**

8.3 Streaming identity

The streaming detector (T6) compares each arm against **its own offline decode** on the same recording, so no ceiling or floor enters and no cross-arm ranking is implied.

The three explicitly causal FocalCodec configurations show near-zero median additional change under the tested encoders, supported recordings and imposed chunking regime. Across all three arms and all three encoders, S1 runs from **0.999866 to 0.999946** and the S2 median additional change runs from -3.9×10^{-5} to -5.3×10^{-6} — small, and consistently negative. That is the whole claim. ⚠️ **This study establishes no minimum detectable streaming change and no equivalence threshold**, so “near-zero median” must not be read as “below the instrument’s resolution”: the calibrated quantity G, the separation between the same-speaker and different-speaker reference distributions, is not a detection limit for a paired within-arm change and is not used as one here. Nothing is claimed to be lost, and nothing is claimed to be preserved.

Every other arm degrades by an amount orders of magnitude larger, from **-0.078** (`bigvgan22` , `ecapa`) to **-0.742** (`dualcodec_25hz_v1` , `ecapa`); S2 is a paired retention delta, not a cosine, and the two are reported separately in T6. For `focalcodec_25hz` under `redimnet_b6_lm` the offline-versus-streamed cosine itself reaches **-0.013**: the streamed output is essentially orthogonal to that arm’s own offline decode *in this embedding space*. That is a statement about speaker-embedding similarity only — it does not establish the absence of waveform, linguistic or spectral relationships, which were not measured here.

This agrees with Gate 4’s Q1 detector, an aligned maximum-error detector on the waveform, which found streamed-versus-offline error at 105–225% of peak for every arm **except** the same three causal configurations at **0.69–0.74%**. Two measurements sharing no implementation, the same three survivors. ⚠️ They also share their input: the same reconstructions of the same corpus. The agreement is **corroborating evidence** — it makes an independent implementation error in both less likely — but agreement between two implementations cannot rule out an error in either, and it says nothing about an artefact of the material both were computed on.

The claim is about streamed behaviour, not defect. Thirteen of these arms were driven with chunked context they were never designed for. What the result establishes is the cost of streaming a decoder that was not built to stream.

9. Audit history, and why it is in the paper

Six rounds of independent adversarial audit ran against this study — four before publication, one against the published package, and one against the correction itself. They are reported

because **three of the defects they found changed conclusions**, and a methods section that hides them would misrepresent how the numbers were obtained.

- **The cross-state reference used pairs the measurement forbade.** Dispersion compared different texts; its human ceiling did not, admitting the *same sentence spoken in two emotions* — in a parallel corpus, the most similar cross-state pair that exists. Excess: 3,255 pairs in Set S. **Eight of 108 cross-state labels changed** when it was repaired.
- **The Set L ceiling was measured on the wrong duration.** Calibration used full-length recordings where the measurement compares 14-second prefixes, inflating the ceiling by 0.020–0.055 cosine against interval widths of 0.02–0.03. Counterfactual: **73 of 108 Set L strata would have carried a different label, 67 of them false drift verdicts.**
- **A pair-grouping convention was order-dependent.** Permuting row order moved cross-state estimates by up to 0.089 — enough to flip a label. Replaced with a symmetric rule and verified invariant to machine precision.

A fourth class of defect was archival rather than scientific: evidence verification hashed the instrument but not the evidence, and the archive validated what it listed rather than what it required. Both are repaired and negative-tested. One superseded snapshot was nonetheless **permanently lost**, and that is recorded as unrecoverable rather than described as retained; the 540 labels were instead reconstructed independently from the unchanged measurement rows, with 0 mismatches.

The pattern is worth naming because it recurred three times in different disguises: **verification that checks what is present rather than what is required.** The estimator without its reference; the instrument without its evidence; the manifest without its inventory.

A fifth round ran after v1.0.0 was published, against the published package rather than the private tree, and it found the same pattern a fourth time — in the release itself. The advertised public reproduction commands did not run: the scripts resolved artifacts relative to their location in the *private* tree, where they sat one directory above the data, and the release ships them flat. The claim audit additionally required a withheld file unconditionally, and the verifier being advertised was the manifest of the private tree, which cannot verify a sanitised package and reported 26 of 32 artifacts as drift. Alongside that, four claims outran their evidence — a first-audio range that borrowed a steady-state number for its upper end, a chunk-failure count of three where two arms fail everywhere and a third fails only at the anchor, a mechanism asserted for Griffin-Lim that this package does not demonstrate, and three sentences claiming more than their evidence could carry — that the causal arms lost *nothing*, when their medians are small but non-zero; that one arm's streamed output bore *no*

relation to its own offline decode, when what was measured was a single embedding distance; and that agreement between two instruments *excluded* an implementation error in either, which agreement cannot do. Every one is corrected in v1.0.1 and itemised in `CORRECTIONS_v1.0.1.md`. **No measurement changed.**

A sixth round then attacked the correction itself, before any of it was published, and found five more defects — including two claims in the repair that outran their evidence, a privacy scanner that still listed roots instead of declaring its domain, and a package verifier that printed its own integrity fingerprint without ever comparing it. Those are corrected too, and recorded in place rather than rewritten. The lesson generalises the one above three times over: a package that verifies its own private origin has not verified what it shipped; a verifier that prints a value has not compared it; and a correction is not exempt from the discipline that produced it.

10. Limitations

1. **Perceptual quality is not measured.** No listening test was run. Which decoder *sounds* best is not established and no sentence here implies it.
2. **The sole ranking metric failed its own validity instrument in Route B** (§7.1), and the check has **not been run** in Route A, which is not the same as passing it.
3. Mel-cepstral distance is gain-invariant by construction and blind to a pure level error; these values are not literature-scale MCD and must never be quoted against published figures.
4. **Q5 measures speaker-encoder representations**, not perceptual identity and not cloning quality.
5. **17,020 of 31,248 offline cells and 26,147 of 27,776 streamed cells fall outside the validated support domain** (14,083 and 781 inside, respectively). Support is carried as a covariate, never as an exclusion, and every aggregate is reported both pooled and stratified.
6. FocalCodec dispersion and band-limiting **cannot be separated** in this design (§8.2).
7. Cross-state markers on two of three encoders are **encoder baseline limitations** (§8.2).
8. Seven speakers, one studio, one chain, English. Absolute values are not comparable to VoxCeleb-scale benchmarks.
9. The incumbent's environment is **not numerically comparable** to the reference environment (§6).
10. `melflow` results rest on **locally repaired upstream code**, and its latency is an eager-mode upper bound.

11. One encoder runs on CPU because it failed an exact-zero cross-process determinism probe on CUDA. The measured CPU-versus-GPU cosine difference is $\leq 2.5e-04$, roughly $100\times$ below the smallest decision bound.

11. Conclusions

For a product decision. If the requirement is genuine streaming, the candidate set is far smaller than the literature suggests: three arms in twenty reproduce full-context output from partial input with load-bearing state, and the same three are the only ones showing **near-zero median additional change under our speaker-embedding metric** when streamed — under the tested encoders, supported recordings and imposed chunking regime. That is a representation-level result measured with three encoders, not a claim that those decoders preserve identity as a listener would judge it, and not a claim that the change is below any established detection threshold, because this study establishes none. Everything else is a chunking strategy whose cost we have now measured rather than assumed.

For measurement practice. Two measurements ranked a zero-parameter phase-blind algorithm first. Neither was wrong about what it measured — a spectral distance and a speaker-embedding similarity, each computed correctly. **Neither establishes perceptual quality**, and this study cannot say what either costs a listener, because no listening test was run. The pre-registration that named this outcome in advance is the only reason it reads as a finding rather than an embarrassment — and the reason no metric was swapped in afterwards to produce a tidier table.

What comes next, and is not claimed here. Human listening and adversarial detectability are declared later phases, neither started. The reconstructions are retained for exactly that reason.

Data and code availability

The analysis code, frozen specifications, machine-readable results, generated tables and the publication manifest are released. **The source audio is not released:** it is private, rights-cleared recordings of seven consented speakers. Derived **speaker embeddings are also withheld** — an embedding of a named human is biometric data, and publishing it is a different act from the research use the consent covers.

What the public package can and cannot do is worth stating exactly, because v1.0.0 of this paper claimed more than it delivered.

Regenerates from the released artifacts. Every table in §§5–8 and every headline number in the prose is recomputed from the published aggregate artifacts by `make_paper_tables.py`, and re-traced to its source artifact by `claim_audit.py`. `make_public_manifest.py --verify` re-hashes the released bytes. All three run from the release package with no private input.

Does not. One claim — the support-domain cell counts, **17,020 of 31,248 offline and 26,147 of 27,776 streamed** — is computed from `Q5_CELLS.jsonl`, the per-recording rows, which are withheld. The public claim audit reports it `WITHHELD`, not `PASS`. More generally, the published aggregates and their confidence intervals **cannot be independently reconstructed from the public package**: that needs the withheld per-recording rows and embeddings. Recomputing a table from a published aggregate is a consistency check, and this paper does not represent it as more than one.

`PUBLICATION_MANIFEST.json` is the frozen manifest of the **private canonical tree** and is published as a historical record; it does not describe the sanitised public package and will report drift against it. `PUBLIC_MANIFEST.json` is the integrity record for the released bytes. The frozen records carry SHA-256 for each artifact, and the private verification tool reports instrument, evidence and archive drift separately, failing closed on any of them. #
GENERATED TABLES — every value read from a frozen artifact

Generated by `paper/make_paper_tables.py`. Publication manifest `1e93e28b9adb3b1f`.

T1 — The speaker-encoder panel (Q5)

slot	encoder	training corpus	device	frozen-study EER (pooled expressive)
E1	<code>redimnet_M_vb2_ptn</code>	VoxBlink2	cuda	0.047 [0.024, 0.072]
E2	<code>ecapa</code>	VoxCeleb2	cuda	0.146 [0.092, 0.177]
E3	<code>redimnet_b6_lm</code>	VoxCeleb2	cpu	0.064 [0.034, 0.082]

T2 — Instrument calibration on SOURCE audio only (Q5 §3, §9)

encoder	G(e) = usable range	tier bound 0.10-G	1-sample-shift response	scale response	F2
<code>ecapa</code>	0.2663	0.0266	0.000561	1.77e-08	OK
<code>redimnet_M_vb2_ptn</code>	0.3824	0.0382	0.000441	8.24e-11	OK
<code>redimnet_b6_lm</code>	0.4056	0.0406	0.000317	0.000726	OK

T3 — Human within-speaker ceiling and between-speaker floor, source audio (Q5 §3)

encoder	set	ceiling $\tilde{C}$ range over states	floor F_{hi} range	separability gate
ecapa	S	0.594 – 0.698	0.291 – 0.447	6/6 PASS
ecapa	L	0.827 – 0.830	0.371 – 0.483	2/2 PASS
redimnet_M_vb2_ptn	S	0.780 – 0.829	0.306 – 0.452	6/6 PASS
redimnet_M_vb2_ptn	L	0.880 – 0.893	0.326 – 0.413	2/2 PASS
redimnet_b6_lm	S	0.724 – 0.746	0.268 – 0.436	6/6 PASS
redimnet_b6_lm	L	0.862 – 0.873	0.303 – 0.500	2/2 PASS

T4 — Speaker-embedding retention, Set S, all six states pooled

— Cosine values are ENCODER-SPECIFIC and are never averaged across encoders.

arm	ecapa	redimnet_M_vb2_ptn	redimnet_b6_lm	spread
griffinlim	0.9952 [0.9950, 0.9954]	0.9955 [0.9954, 0.9957]	0.9957 [0.9954, 0.9958]	0.0004
bigvgan22	0.9936 [0.9934, 0.9938]	0.9933 [0.9932, 0.9937]	0.9911 [0.9907, 0.9913]	0.0024
vocos_mel24	0.9878 [0.9874, 0.9882]	0.9837 [0.9829, 0.9841]	0.9821 [0.9816, 0.9829]	0.0057
qwen3_tts_tokenizer_12hz	0.9417 [0.9391, 0.9427]	0.9253 [0.9237, 0.9277]	0.9082 [0.9067, 0.9115]	0.0335
dac44	0.9324 [0.9283, 0.9342]	0.9191 [0.9154, 0.9207]	0.8972 [0.8934, 0.8999]	0.0352
dualcodec_25hz_v1	0.9242 [0.9214, 0.9269]	0.9048 [0.9021, 0.9073]	0.8876 [0.8843, 0.8908]	0.0366
fish_modified_dac	0.9229 [0.9207, 0.9259]	0.8941 [0.8912, 0.8970]	0.8764 [0.8726, 0.8800]	0.0466
encodec_vocos	0.9146 [0.9121, 0.9180]	0.8800 [0.8754, 0.8838]	0.8546 [0.8520, 0.8593]	0.0601
mimi_q32	0.9073 [0.9027, 0.9093]	0.8653 [0.8613, 0.8686]	0.8581 [0.8533, 0.8622]	0.0492
encodec24_q8	0.8781 [0.8721, 0.8812]	0.7588 [0.7534, 0.7664]	0.8003 [0.7927, 0.8106]	0.1193

arm	ecapa	redimnet_M_vb2_ptn	redimnet_b6_lm	spread
dualcodec_12hz_v1	0.8621 [0.8580, 0.8658]	0.8222 [0.8157, 0.8251]	0.8033 [0.7932, 0.8048]	0.0588
mimi_q8	0.7485 [0.7397, 0.7567]	0.6647 [0.6586, 0.6753]	0.6576 [0.6401, 0.6641]	0.0909
focalcodec_50hz	0.6833 [0.6765, 0.6923]	0.6300 [0.6253, 0.6412]	0.6361 [0.6284, 0.6447]	0.0533
focalcodec_50hz_65k_causal	0.6698 [0.6635, 0.6772]	0.6317 [0.6216, 0.6397]	0.6009 [0.5902, 0.6082]	0.0689
focalcodec_50hz_4k_causal	0.5995 [0.5962, 0.6115]	0.5613 [0.5509, 0.5720]	0.5236 [0.5181, 0.5362]	0.0759
focalcodec_50hz_2k_causal	0.5835 [0.5779, 0.6033]	0.5578 [0.5463, 0.5666]	0.5322 [0.5154, 0.5393]	0.0512
focalcodec_25hz	0.5822 [0.5668, 0.5857]	0.5347 [0.5320, 0.5496]	0.5358 [0.5205, 0.5458]	0.0475
focalcodec_12_5hz	0.4316 [0.4141, 0.4379]	0.4287 [0.4210, 0.4398]	0.3936 [0.3773, 0.4039]	0.0380

T5 — Q5 identity labels, 540 strata (18 arms × 3 encoders × 2 sets × strata)

label	count
NOT ESTABLISHED	314
INCONSISTENT / DRIFTING	152
CROSS-STATE DRIFTING	24
CROSS-STATE CONSISTENT	23
PRESERVED + CONSISTENT	19
STABLE OFFSET	8
total	540

Encoder agreement over 180 (arm × set × stratum) comparisons: 1/3 9 · 2/3 87 · 3/3 84. Dissent counts: redimnet_b6_lm 53 · redimnet_M_vb2_ptn 28 · ecapa 24.

T6 — Streaming identity detector (Q5 §7). Within-arm; never a ranking

S1 = cos(offline, streamed) on the same recording. S2 = retention(streamed) – retention(offline), paired.

arm	S1 ecapa	S2 ecapa	S1 redimnet_M_vb2_ptn	S2 redimnet_M_vb2_ptn	n paired
focalcodec_50hz_2k_causal	0.9999	-0.0000	0.9999	-0.0000	1680
focalcodec_50hz_65k_causal	0.9999	-0.0000	0.9999	-0.0000	1680
focalcodec_50hz_4k_causal	0.9999	-0.0000	0.9999	-0.0000	1680
bigvgan22	0.9153	-0.0778	0.8861	-0.1148	1624
encodec_vocos	0.7961	-0.1461	0.7643	-0.2001	1680
mimi_q8	0.7432	-0.1481	0.7683	-0.1580	1680
mimi_q32	0.8154	-0.1508	0.8445	-0.1468	1680
focalcodec_12_5hz	0.4808	-0.1859	0.6099	-0.1670	1624
encodec24_q8	0.7051	-0.2329	0.7421	-0.1673	1680
qwen3_tts_tokenizer_12hz	0.6636	-0.2707	0.6625	-0.2805	1680
dac44	0.6333	-0.3379	0.6037	-0.3883	1624
fish_modified_dac	0.5196	-0.4054	0.5404	-0.4321	1680
focalcodec_25hz	0.0754	-0.5629	0.0961	-0.4975	1680
dualcodec_12hz_v1	0.3305	-0.5830	0.4415	-0.5283	1624
focalcodec_50hz	0.0233	-0.6688	0.1646	-0.5444	1624
dualcodec_25hz_v1	0.2008	-0.7420	0.3068	-0.6804	1624
vocos_mel24	—	—	—	—	NOT MEASURED — no streamed output
griffinlim	—	—	—	—	NOT MEASURED — no streamed output

T7 — Measurement volume and admissibility

quantity	value
Q5 measurement rows	177072

quantity	value
speaker embeddings retained	188958
offline cells per arm	1,736 (Set S 1,302 · Set L 210 · ladder 224)
distinct source recordings	1512
speakers	7 (English, one studio, one chain)
Q5 labels	540
cross-state pair counts, reference = estimator	Set S 97,650 · Set L 1,470

T-Q9 — The metric-validity instrument (Gate 4 §5, Route B)

Pre-registered consequence, written before any quality number existed: A metric on which a 32-iteration Griffin-Lim reconstruction scores close to a trained neural codec is a metric that is not resolving what Gate 4 claims to resolve, and that is a finding about the metric.

Verdict:  Q9 FIRES. THE SOLE PREREGISTERED CROSS-ARM RANKING METRIC FAILED ITS OWN VALIDITY CHECK IN ROUTE B.

Per-state values are in `GATE4_STAGE2_TABLES.md` §4, generated from the same artifact.

Gate 4 pairwise verdicts (108 within-route pairs): {“1_metric_validity”: “The benchmark ran correctly and its sole preregistered cross-arm quality ruler FAILED ITS OWN PREREGISTERED VALIDITY CHECK in Route B (Q9). See metric_validity.”,
“2_the_robust_quality_claim”: {“claim”: “mimi_q8 is worse than 3 named arms under the strongest evidence class (SPEC S7.3 ROBUST)”, “arms”: [“dac44”, “fish_modified_dac”, “qwen3_tts_tokenizer_12hz”], “basis”: “n_pa